

## **FMUSER FUQ-200B Multiplexer Scrambler Modulator 112 DVB-C QAM Channels 10G SFP+-RJ45 IP Input-Output Web-Based Management Broadcast Cable TV** Product Overview



The FMUSER FUQ-200B integrates a multiplexer, scrambler, and modulator, featuring 10G SFP+/SFP ports and GE ports for IP input/output. It processes 10G optical signals efficiently, supporting up to 112 DVB-C QAM channels. Housed in a compact 1U chassis, it offers web-based management for easy control. Ideal for broadcast systems, enabling high-density operations in cable TV and digital broadcasting. The device is equipped with one fixed 64ch QAM card and another card slot to expand qam card (16/32/48QAM optional). Its high density helps operators to save cost to the most extent.

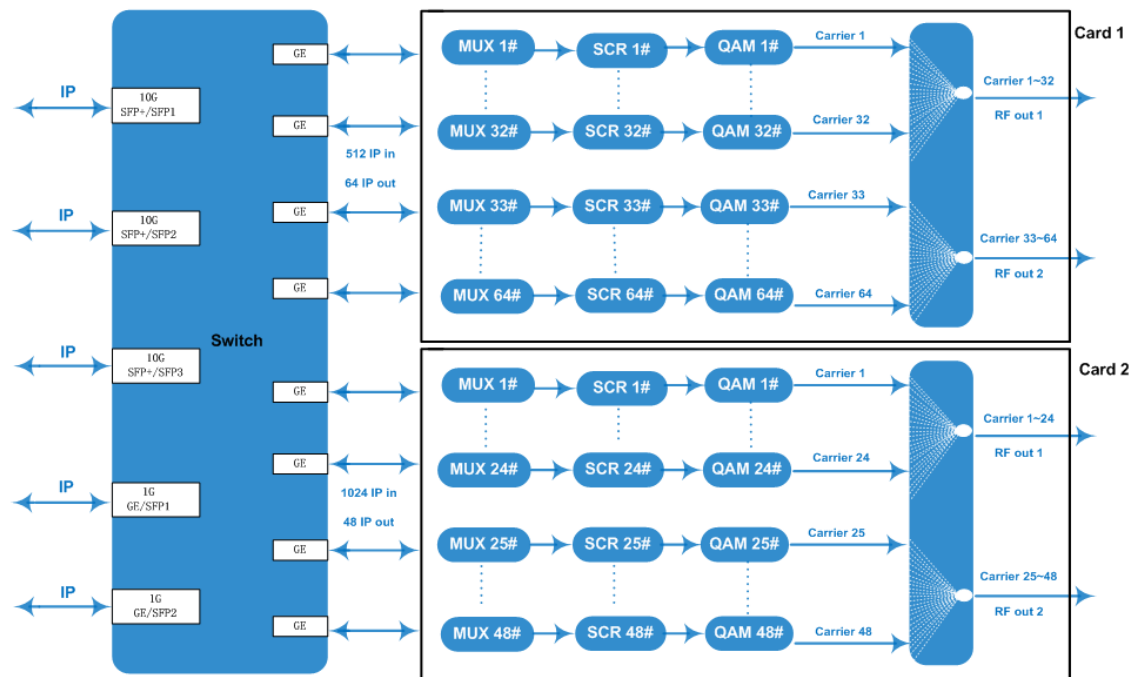
### Key feature

Supports high-density modular design in a 1U chassis with plug-in QAM cards, accommodating up to 112 non-adjacent carriers.

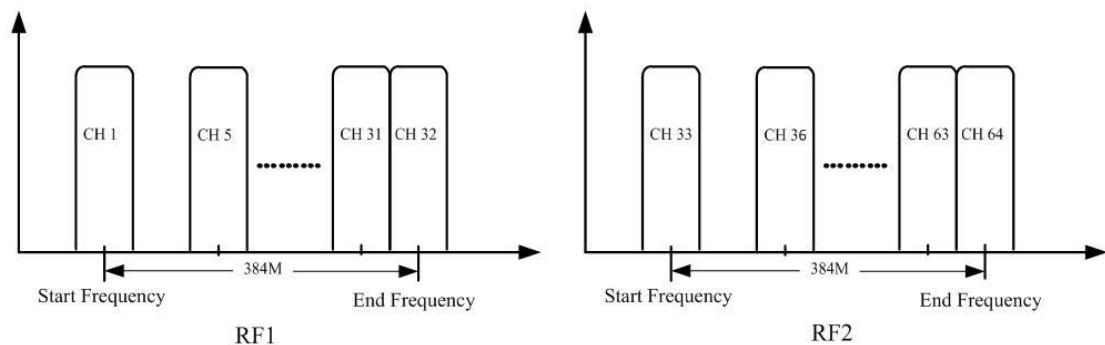
Features 10G SFP+/SFP ports and GE ports for flexible IP input/output, handling up to 9600Mbps per channel.

Integrates multiplexer and scrambler with 6 CAS Simul-cryption, PID remapping, and PSI/SI editing. Enables web-based management for easy network control and configuration.

Inner Principle Chart (Example: one 64qam card + one 48qam card)



Carrier Setting Illustration (Example: 64qam card)



#### Specifications

|       |                          |                                                   |                                                   |                                                   |
|-------|--------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| Input | Input from Front Panel   | Max 512 IP in per card through SFP+/SFP ports 1-3 | Max 512 IP in per card through SFP+/SFP ports 1-3 | Max 512 IP in per card through SFP+/SFP ports 1-3 |
| Input | Input from Daughter Card | IP from 2 GE ports (SFP port optional)            | IP from 2 GE ports (SFP port optional)            | IP from 2 GE ports (SFP port optional)            |
| Input | Transmission Rate        | Max 9600Mbps for each 10G SFP+/SFP input          | Max 9600Mbps for each 10G SFP+/SFP input          | Max 9600Mbps for each 10G SFP+/SFP input          |

| Input      | Transport Protocol  | TS over UDP/RTP, unicast and multicast, IGMP V2/V3 | TS over UDP/RTP, unicast and multicast, IGMP V2/V3                                         | TS over UDP/RTP, unicast and multicast, IGMP V2/V3                                         |
|------------|---------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Mux        | Max PIDs Remapping  | 256 per output channel                             | 256 per output channel                                                                     | 256 per output channel                                                                     |
| Mux        | Functions           | PID remapping (auto/manually optional)             | PID remapping (auto/manually optional)                                                     | PID remapping (auto/manually optional)                                                     |
| Mux        | Functions           | PCR accurate adjusting                             | PCR accurate adjusting                                                                     | PCR accurate adjusting                                                                     |
| Mux        | Functions           | PSI/SI table automatically generating              | PSI/SI table automatically generating                                                      | PSI/SI table automatically generating                                                      |
| Scrambling | Max simulcrypt CA   | Max simulcrypt CA                                  | 6                                                                                          | 6                                                                                          |
| Scrambling | Scramble Standard   | Scramble Standard                                  | ETR289, ETSI 101 197, ETSI 103 197                                                         | ETR289, ETSI 101 197, ETSI 103 197                                                         |
| Scrambling | Connection          | Connection                                         | Local/remote connection                                                                    | Local/remote connection                                                                    |
| Modulation | Modulation Standard | Modulation Standard                                | EN300 429/ITU-T J.83A/B/C                                                                  | EN300 429/ITU-T J.83A/B/C                                                                  |
| Modulation | Constellation       | Constellation                                      | J.83A                                                                                      | Constellation :16/32/64/128/256QAM                                                         |
| Modulation | Constellation       | Constellation                                      | J.83A                                                                                      | Bandwidth :8M                                                                              |
| Modulation | Constellation       | Constellation                                      | J.83B/C                                                                                    | Constellation :64/256QAM                                                                   |
| Modulation | Constellation       | Constellation                                      | J.83B/C                                                                                    | Bandwidth :6M                                                                              |
| Modulation | QAM Channel         | QAM Channel                                        | 32/48/64/80/96/112 non-adjacent carriers output, 192 or 384Mbps bandwidth for each RF port | 32/48/64/80/96/112 non-adjacent carriers output, 192 or 384Mbps bandwidth for each RF port |

|            |                                    |                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                         |
|------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modulation | Symbol Rate                        | Symbol Rate                                                                                                                                                                                                                                                                             | 3600~7000Ksps,<br>1ksps stepping<br>5057Ksps<br>(J.83B, 64QAM) ;<br>5361Ksps<br>(J.83B, 256QAM)                                                                                                                                                                                         | 3600~7000Ksps,<br>1ksps stepping<br>5057Ksps<br>(J.83B, 64QAM) ;<br>5361Ksps<br>(J.83B, 256QAM)                                                                                                                                                                                         |
| Modulation | Constellation                      | Constellation                                                                                                                                                                                                                                                                           | 16, 32, 64, 128,<br>256QAM                                                                                                                                                                                                                                                              | 16, 32, 64, 128,<br>256QAM                                                                                                                                                                                                                                                              |
| Modulation | FEC                                | FEC                                                                                                                                                                                                                                                                                     | RS (204, 188)                                                                                                                                                                                                                                                                           | RS (204, 188)                                                                                                                                                                                                                                                                           |
| RF Output  | Interface (per<br>card)            | 1 or 2 F type<br>output ports for<br>16/32/48/64<br>carriers card,<br>75Ω. 16 or 32 ch<br>qam card : all<br>Carrier out thru<br>one RF port 48ch<br>qam card :<br>Carrier 1~24 out<br>thru RF1, 25~48<br>thru RF2 64ch<br>qam card :<br>Carrier 1~32 out<br>thru RF1, 33~64<br>thru RF2 | 1 or 2 F type<br>output ports for<br>16/32/48/64<br>carriers card,<br>75Ω. 16 or 32 ch<br>qam card : all<br>Carrier out thru<br>one RF port 48ch<br>qam card :<br>Carrier 1~24 out<br>thru RF1, 25~48<br>thru RF2 64ch<br>qam card :<br>Carrier 1~32 out<br>thru RF1, 33~64<br>thru RF2 | 1 or 2 F type<br>output ports for<br>16/32/48/64<br>carriers card,<br>75Ω. 16 or 32 ch<br>qam card : all<br>Carrier out thru<br>one RF port 48ch<br>qam card :<br>Carrier 1~24 out<br>thru RF1, 25~48<br>thru RF2 64ch<br>qam card :<br>Carrier 1~32 out<br>thru RF1, 33~64<br>thru RF2 |
| RF Output  | RF Range                           | 50~960MHz,<br>1kHz stepping                                                                                                                                                                                                                                                             | 50~960MHz,<br>1kHz stepping                                                                                                                                                                                                                                                             | 50~960MHz,<br>1kHz stepping                                                                                                                                                                                                                                                             |
| RF Output  | Output Level                       | -20dBm~+10dB<br>m(87~117dbμV),<br>0.1dB stepping                                                                                                                                                                                                                                        | -20dBm~+10dB<br>m(87~117dbμV),<br>0.1dB stepping                                                                                                                                                                                                                                        | -20dBm~+10dB<br>m(87~117dbμV),<br>0.1dB stepping                                                                                                                                                                                                                                        |
| RF Output  | MER                                | ≥ 40dB                                                                                                                                                                                                                                                                                  | ≥ 40dB                                                                                                                                                                                                                                                                                  | ≥ 40dB                                                                                                                                                                                                                                                                                  |
| TS output  | Per Daughter<br>QAM Card           | 16/32/48/64 IP<br>output over<br>UDP/RTP/RTSP,<br>unicast/multicast,<br>through<br>SFP+/SFP ports<br>1-3 or GE port<br>RJ45                                                                                                                                                             | 16/32/48/64 IP<br>output over<br>UDP/RTP/RTSP,<br>unicast/multicast,<br>through<br>SFP+/SFP ports<br>1-3 or GE port<br>RJ45                                                                                                                                                             | 16/32/48/64 IP<br>output over<br>UDP/RTP/RTSP,<br>unicast/multicast,<br>through<br>SFP+/SFP ports<br>1-3 or GE port<br>RJ45                                                                                                                                                             |
| System     | Web-based<br>Network<br>management | Web-based<br>Network<br>management                                                                                                                                                                                                                                                      | Web-based<br>Network<br>management                                                                                                                                                                                                                                                      | Web-based<br>Network<br>management                                                                                                                                                                                                                                                      |

|         |              |                                                                |                                                                |                                                                |
|---------|--------------|----------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|
| General | Dimension    | 420mm×440mm<br>×44.5mm<br>(W×L×H)                              | 420mm×440mm<br>×44.5mm<br>(W×L×H)                              | 420mm×440mm<br>×44.5mm<br>(W×L×H)                              |
| General | Temperature  | 0~45■(operation<br>, -20~80■(storage)                          | 0~45■(operation<br>, -20~80■(storage)                          | 0~45■(operation<br>, -20~80■(storage)                          |
| General | Power Supply | AC 100V±10%,<br>50/60Hz ; or AC<br>220V±10%,<br>50/60Hz        | AC 100V±10%,<br>50/60Hz ; or AC<br>220V±10%,<br>50/60Hz        | AC 100V±10%,<br>50/60Hz ; or AC<br>220V±10%,<br>50/60Hz        |
| General | Consumption  | 50W(1 daughter<br>QAM<br>card)/75W(2<br>daughter QAM<br>cards) | 50W(1 daughter<br>QAM<br>card)/75W(2<br>daughter QAM<br>cards) | 50W(1 daughter<br>QAM<br>card)/75W(2<br>daughter QAM<br>cards) |

#### Order Guide

| Options Available | Combinations                   |
|-------------------|--------------------------------|
| 32ch QAM out      | 32qam card * 1                 |
| 48ch QAM out      | 48qam card * 1                 |
| 64ch QAM out      | 64qam card * 1                 |
| 80ch QAM out      | 64qam card * 1 + 16qam card *1 |
| 96ch QAM out      | 64qam card * 1 + 32qam card *1 |
| 112ch QAM out     | 64qam card * 1 + 48qam card *1 |

#### About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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